

Anakin Dey

Curriculum Vitæ

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Research Interests

Algebraic Combinatorics; Algebraic Geometry and Commutative Algebra; Computational Algebra; Algebraic and Fine-Grained Complexity; Algebraic Methods in Combinatorics; Graph Theory; Combinatorial Optimization;

Education

2020–2024 **BS in Mathematics**, *University of Illinois Urbana-Champaign*, GPA: 3.94 / 4.00

Publications

Classifying Primitive Solvable Permutation Groups of Rank 5 and 6

Anakin Dey, Kolton O'Neal, Duc Van Khanh Tran, Camron Upshur, and Yong Yang
Submitted, [arXiv:2308.13043](https://arxiv.org/abs/2308.13043)

Post-Disaster Repair Crew Assignment Optimization Using Minimum Latency

Anakin Dey and Melkior Ornik
IEEE International Smart Cities Conference 2022, [arXiv:2206.00597](https://arxiv.org/abs/2206.00597)

Invited Talks

2023 **Classifying Primitive Solvable Permutation Groups of Rank 5 and 6**

UIUC Department of Mathematics Undergraduate Friday Seminars

2022 **Post-Disaster Repair Crew Assignment Optimization Using Minimum Latency**

IEEE International Smart Cities Conference 2022

Research Experience

Summer 2023 **Undergraduate Research Assistant**, [REU in Algebra, Combinatorics, and Statistics](#),
Under Prof. Yong Yang, Texas State University

Fall 2021 – **Undergraduate Research Assistant**, [LEADCAT Lab](#), Under Prof. Melkior Ornik,
Present University of Illinois Urbana-Champaign

Student Organizations

Founder and President, ACM [SIGma](#)

Special Interest Group in Math and Algorithms

Admin, ACM [SIGPwny](#)

Special Interest Group in Cybersecurity

Teaching Experience

Spring 2021 – Present **Undergraduate Course Assistant**, University of Illinois Urbana-Champaign

- CS 125 - Introduction to Computer Science
Spring 2021 Course Assistant
- CS 173 - Discrete Structures
Spring 2021 - Spring 2022 Course Assistant
Fall 2022 – Present Head Course Assistant
- CS 475 - Models of Computation
Spring 2023 Course Assistant
- CS 374 - Introduction to Algorithms and Models of Computation
Spring 2022 - Fall 2022 Course Assistant
Fall 2023 – Present Course Assistant

Professional Experience

Summer 2022 **Software Engineering Intern**, CME Group, Chicago, IL

Summer 2021 **Software Engineering Intern**, SiteIQ, Lombard, IL

Selected Coursework

- Math 500 Abstract Algebra I
Dummit and Foote - Abstract Algebra
- Math 501 Abstract Algebra II
Joseph J. Rotman - Advanced Modern Algebra Parts 1 and 2
- Math 502 Commutative Algebra
Hideyuki Matsumura - Commutative Ring Theory
- Math 511 Introduction to Algebraic Geometry
Igor Shafarevich - Basic Algebraic Geometry, Volume 1
- Math 518 Differentiable Manifolds I
John M. Lee - Introduction to Smooth Manifolds
- Math 581 Extremal Graph Theory
Douglas B. West - The Art of Combinatorics Volume I: Extremal Graph Theory
- CS 473 Algorithms
Jeff Erickson - Algorithms
- CS 475 Models of Computation
Michael Sipser - Introduction to the Theory of Computation
- CS 598TMC Algorithms from the Fine-Grained Perspective
- CS 598MAF Algebraic and Geometric Complexity Theory
- CS 598TH1 Advances in Theoretical CS